

NFL Defensive Coverage Predictor

CS/SE 4AL3 – Project Proposal

Group 77

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Task Overview and Significance

Our task is to recognize defensive coverages before the snap. This is challenging because defenses often try to disguise their coverage by showing one formation or alignment before the snap and executing another after the snap. This makes it difficult for an offensive player or coordinator to determine the defense's true coverage based only on its pre-snap alignment.

This matters because accurately identifying defensive coverage before the play begins could help offensive coordinators and quarterbacks make better pre-snap decisions, such as adjusting the play, route combinations, or blocking scheme.

This project is non-trivial because the model must make its prediction using only pre-snap information, while accounting for defensive disguises, player positioning, and different formations. The novel aspect of our project is developing a machine learning approach specifically focused on predicting the coverage before the snap, rather than using information from the play after it has already developed. We will also investigate which pre-snap features are most useful for distinguishing between different defensive coverages.

Task Definition

Input/data type: [e.g., text, images, tabular records]

Prediction task: [classification / regression / generation]

Labels and output: [number of classes; label definitions; single-label or multi-label; or regression target]

Unit of prediction: [What one data point represents]

[State the task precisely, including the model inputs and desired outputs.]

Data Sources and Collection Plan

We downloaded the [Big Data Bowl regional-event dataset](#): 122 games from weeks 1–8 of the 2021 season. Its documentation identifies NFL Next Gen Stats as the tracking provider and Pro Football Focus (PFF) as the scouting and coverage-label provider. [Next Gen Stats](#) collects player movement through stadium receivers and RFID tags at 10 samples per second. The [NFL](#) organizes the release; detailed PFF annotation procedures are not documented.

We obtained CSVs directly from GitHub after the original [Kaggle 2025 dataset](#) was removed. No scraping or API polling is needed. Raw data stays in the ignored `data/regional-2021/` directory; a manifest records the source revision and file hashes. Data-use and redistribution terms remain to be verified.

The release contains game, play, player, PFF scouting, and per-game tracking tables. We join plays and tracking on `gameId/playId`, and player metadata on `nflId`. Available metadata includes week, teams, player positions, timestamps, direction, and snap events. Inputs will combine player coordinates, speed and acceleration with down, yards to go, clock, and pre-snap scores. The existing `pff_passCoverage` labels provide 12 observed classes; no new annotation is required.

We will retain the last complete frame strictly before each play’s `ball_snap`, with 11 distinct players per team and valid coordinates, then normalize direction and positions relative to the line of scrimmage. One snapshot per play keeps the input small while preserving alignment. We exclude coverage annotations, post-snap frames, descriptions, and outcomes from predictors. Games stay together across splits; preprocessing is fitted on training data only.

Dataset size: There are 8,557 plays and 8,314,178 tracking rows. The initial audit retained 8,522 labeled plays with complete pre-snap snapshots and excluded 35 that failed these checks. Plays, rather than frames, count as instances; further quality checks may reduce this total.

Three actual labeled examples: Inputs contain all 22 player coordinates and game context. Below are compact summaries of three source records: down/distance, defensive coordinate ranges in yards, and the recorded PFF label.

1. Game 2021090900, play 97, frame 5: down 3, 2 yards to go; defensive x range 43.38–58.88, y range

- 15.98–37.61 yards. **Label:** Cover-1.
2. Game 2021090900, play 137, frame 6: down 1, 10 yards to go; defensive x range 91.82–108.55, y range 8.77–43.75 yards. **Label:** Cover-3.
3. Game 2021090900, play 456, frame 5: down 2, 10 yards to go; defensive x range 31.23–45.91, y range 10.47–44.64 yards. **Label:** Cover-6.

Proposed Solution and Evaluation

[Describe features and target labels, candidate models, and how you will approach the problem. Do not propose simple linear regression. Discuss existing solutions and how your work differs. Explain evaluation design, metrics, data splits/baselines, and how you will decide whether the model is good. Name intended libraries.]

Planned libraries: [e.g., scikit-learn, PyTorch, pandas]

Evaluation: [Metrics, baseline(s), validation plan, and success criteria]

Sources inspiring the approach (2–5)

1. [Author(s), title, venue/year, URL or DOI]
2. [Author(s), title, venue/year, URL or DOI]
3. [Optional source]

TA Consultation

We consulted with Abdelaziz Serour twice throughout the development of our project idea. During our first interaction, our team proposed developing a SpecDec system for tool calls. However, after discussing the idea with the TA, we concluded that it was too complex and theoretical for the scope of the project. As a result, we decided to pivot toward a more practical and applied idea.

We then developed the idea of recognizing defensive coverages during the pre-snap period. We reached out to Abdelaziz again through Outlook to discuss whether this idea was feasible, and he agreed that it was a suitable project that we could proceed with.

Team Contract

Team mission: [What is the purpose of the team?]

Roles, duties, and expectations: [Describe each member’s responsibilities and shared expectations. Focus on how you will work together.]

Leadership and coordination: [How will you organize meetings, make decisions, track work, and share facilitation?]

Communication and conflict resolution: [How will you raise concerns, meet, discuss disagreements, and resolve conflict? Add any other useful ground rules.]

Signatures

Printed name	Signature	Date
Justin Kwinecki	_____	October 7, 2026
Zayed Umer	_____	October 7, 2026
William Pol	_____	October 7, 2026